IDEATION PHASE

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Problem Identification

In many applications, users are given access to records based only on their roles or general permissions. However, some records may contain sensitive or restricted information that should be accessible only under specific conditions. Role-based access alone may not provide sufficient control over individual records.

Therefore, a solution is needed to restrict record access based on specific field values and user conditions.

2. Proposed Idea

The proposed project, Script-Controlled ACL – Restrict Record Access Based on Field Value, uses a custom script within an Access Control List (ACL) to control access to records.

The ACL script checks the values of specific fields in a record and verifies the conditions related to the current user. Access is granted only when the defined conditions are satisfied and the script evaluates to true.

3. Objectives

• To provide secure record-level access control.
• To restrict records based on specific field values.
• To verify user conditions before granting access.
• To reduce unauthorized access to sensitive records.
• To provide flexible and customizable access rules using scripts.

4. Key Features

• Field-Based Access: Records can be restricted based on field values.
• User-Based Validation: The current user's conditions can be checked.
• Script-Controlled Security: Custom scripts determine whether access is allowed.
• Record-Level Restriction: Access can be controlled for individual records.
• Flexible Rules: Different conditions can be created based on application requirements.

5. Expected Outcome

The system will provide controlled access to records by evaluating predefined field and user conditions through an ACL script. If the conditions are satisfied, the user can access the record; otherwise, access will be denied.

This approach improves record security and provides a flexible mechanism for implementing customized access-control requirements.